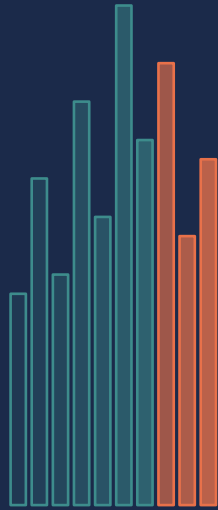


Analyzing Death Age Differences Between Right-Handers and Left-Handers

A Bayesian Statistics Approach



The Question

Do left-handers really die younger?

The Original Claim

In 1991, Halpern & Coren reported that left-handers died on average **9 years earlier** than right-handers — a finding that received widespread media coverage and sparked significant scientific debate.



The claim was never conclusively proven or disproven — until a Bayesian approach reveals the true explanation.

This Analysis

Dataset 1

National Geographic Survey (1986) — left-handedness rates by age & gender

Dataset 2

CDC US Death Distribution (1999) — deaths by age, both sexes

Method

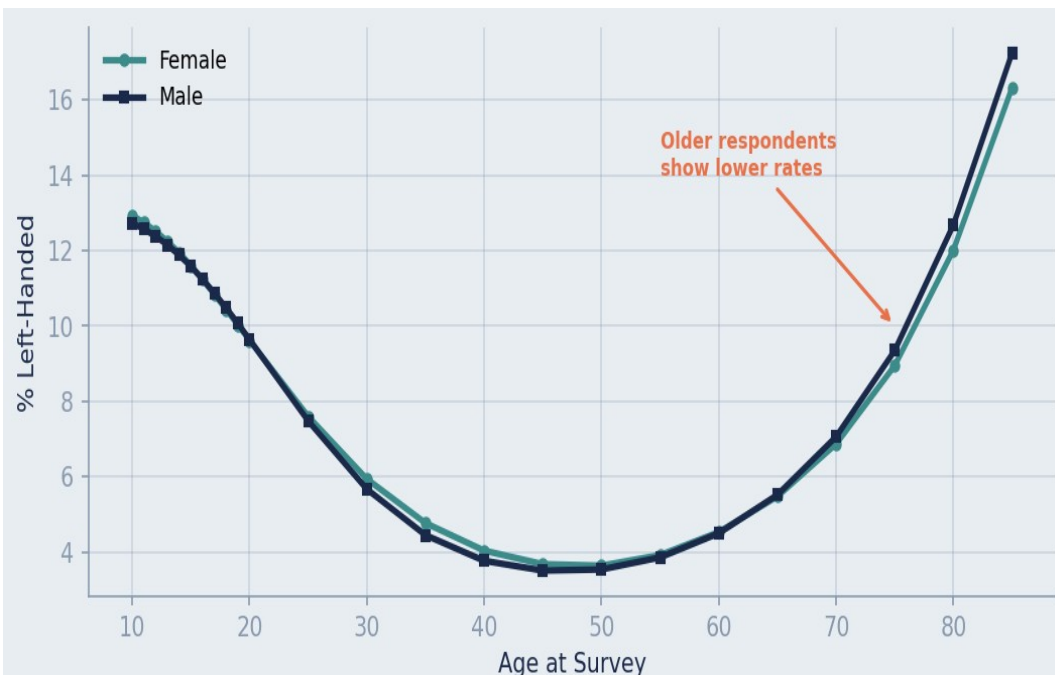
Bayesian inference: $P(A \mid \text{Handedness})$ via Bayes' Theorem

Tool

Python — pandas, numpy, matplotlib

The Data: Handedness Rates by Age

National Geographic Survey, 1986 • 76 age groups • Male & Female



~13%

Left-handedness rate
for those born 1960s+

~3.5%

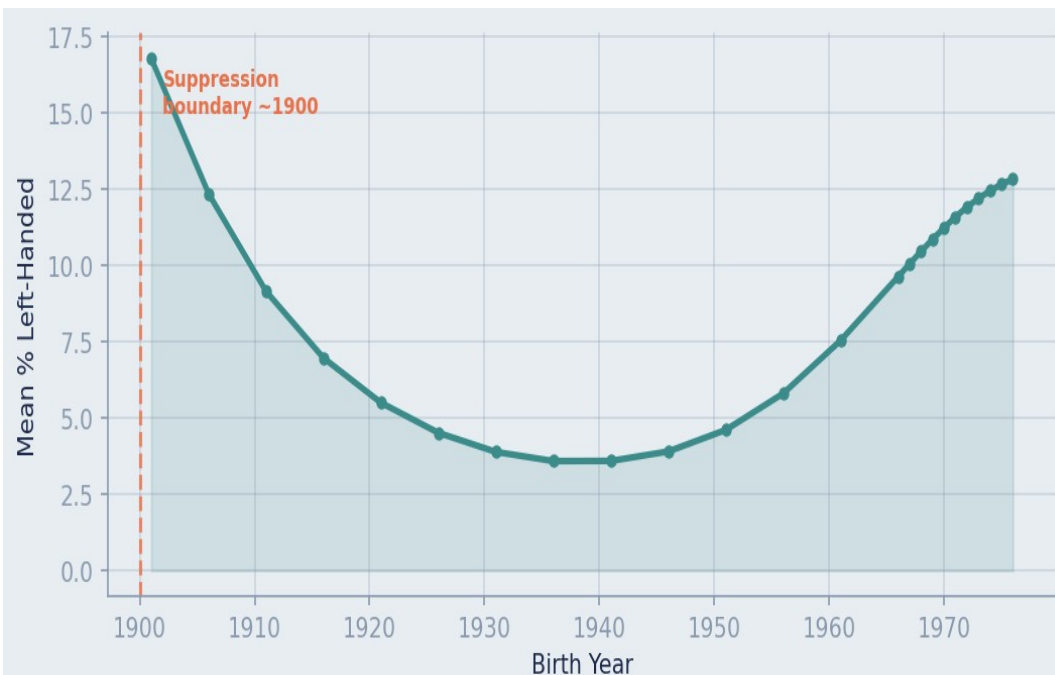
Left-handedness rate
for those born pre-1900

Key

This gap is NOT about
mortality — it's social history

The Cohort Effect: A Generational Story

When re-expressed by birth year, the social suppression of left-handedness becomes undeniable



Timeline of Change



Pre-1900

Strong societal pressure; natural left-handers forced to use right hand. Reported rates: ~3.5%



1900-1940

Gradual relaxation of strictness in schools and households. Rates begin rising slowly.



1940-1970

Post-war cultural shifts. Left-handedness increasingly accepted. Rates climb to 10%+



1970s+

Full acceptance. Natural rate of ~13% finally reflected in surveys.

The Bayesian Framework

Applying Bayes' Theorem to compute $P(\text{Age at Death} \mid \text{Handedness})$

$$P(A \mid LH) = P(LH \mid A) \times P(A) / P(LH)$$



$P(LH \mid A)$

Probability of being left-handed given age at death. Derived from 1986 survey cohort rates. Low for old ages (suppression era), higher for modern birth cohorts.



$P(A)$

Probability of dying at age A. Taken directly from CDC US death distribution data — the base mortality rate at each age for the population.

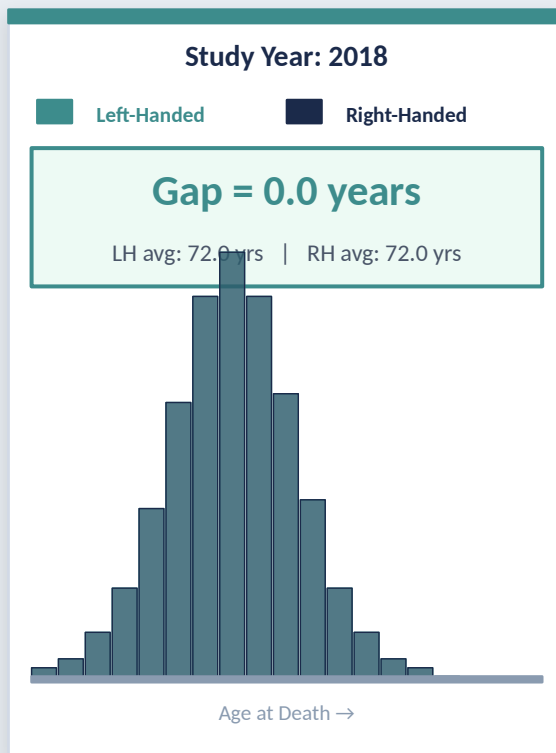
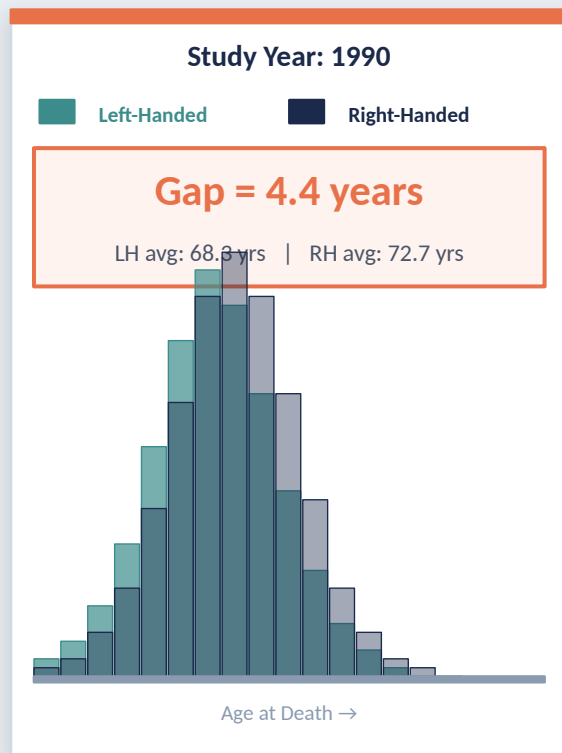


$P(LH)$

Overall probability of left-handedness in the deceased population. Computed via the Law of Total Probability: $\sum P(LH \mid A) \times N(A) / \sum N(A)$.

Results: The Gap That Disappears

Comparing posterior death-age distributions between 1990 and 2018 reveals the statistical artifact



Key Numbers

Year	LH Avg	RH Avg	Gap
1990	68.3	72.7	4.4 yrs
2018	72.0	72.0	0.0 yrs

The gap shrinks
from
4.4 yrs to **0.0 yrs**

as suppression-era
generations leave
the mortality data.

Conclusion

Left-handers do not die younger — the claim is a statistical artifact



The Cohort Effect

Before ~1940, social pressure suppressed left-handedness reporting. Elderly deceased had artificially low LH rates — not because LH died young.



Bayes Explains It

Conditioning on being LH in 1990 statistically selected for younger birth cohorts, shifting the apparent death distribution leftward.



Time Proves It

By 2018 the gap closes to exactly 0.0 years, as post-suppression generations enter the mortality data — confirming the artifact.



The Verdict

No biological mechanism causes left-handers to die earlier. The 1991 finding was entirely a product of changing social norms across generations.